

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
1	(a)		Award 1 mark for ± 0.1 [cm] or measure to resolution of ruler Award 2 marks for ± 0.2 [cm]	1	1		2	1	2
	(b)	(i)	$V = \pi \left(\frac{d}{2} \right)^2 l = \pi \left(\frac{1.5 \times 10^{-3}}{2} \right)^2 \times 11.5 \times 10^{-2}$ (subst and convincing change of units for l and d) (1) $= 2.03 \times 10^{-7} \text{ m}^3$ or 0.2 cm^3 or 203 mm^3 (1) unit mark	1	1		2	2	2
		(ii)	$p_V = 2p_d + p_l = 2 \frac{0.1}{1.5} (1) + \frac{0.2}{11.5} (1)$ (or by impl.) $= 0.133 + 0.017 = 0.150$ (1) allow ecf for this mark, on the use of 0.1 instead of 0.2 only Hence $p_V = 15\%$ (Alternative: use percentages throughout)	1	1 1		3	3	3
	(c)	(i)	From intercepts with x -axis, mean = -270 [$^{\circ}\text{C}$] (1) Uncertainty = 20 [$^{\circ}\text{C}$] (1) Award 1 mark only if 270 used		2		2	1	2
		(ii)	Any 4 $\times$ (1) from: -Straight line -Intercept is consistent i.e. -273 $^{\circ}\text{C}$ / line would go through the origin if temp plotted in K -Passes through all error bars -Volume linked to length - $V \propto T$ or $l \propto T$			4	4	1	4
	(d)		Kinetic or internal energy (or velocity) approaches minimum / zero. Accept very little KE / stopped moving / molecules stop / stops vibrating Don't accept KE decreases greatly / superconduct or superfluid	1			1		